

Multiplying Fractions and Mixed Numbers

Answer Key

Grade 4–5 Fractions

Quick Answers

1. $\frac{2}{5}$
 2. $\frac{3}{8}$
 3. $\frac{10}{3}$
 4. $\frac{1}{12}$

5. $\frac{3}{5}$
 6. $\frac{3}{5}$
 7. $\frac{3}{5}$
 8. $\frac{9}{2}$

9. $\frac{1}{4}$
 10. 4.125
 11. $\frac{8}{3}$
 12. $\frac{3}{4}$

Common wrong answers

10. *If they got 2.375: multiplied whole parts and fraction parts separately instead of converting to improper fractions*

1. $\frac{2}{3} \times \frac{3}{5}$. Multiply straight across: numerator $2 \times 3 = 6$, denominator $3 \times 5 = 15$, giving $\frac{6}{15}$. Both parts share the factor 3, so divide top and bottom by 3. $\boxed{\frac{2}{5}}$

2. $\frac{3}{4} \times \frac{1}{2}$. Across the top: $3 \times 1 = 3$. Across the bottom: $4 \times 2 = 8$. The numerator 3 and the denominator 8 share no factor, so it is already simplest. $\boxed{\frac{3}{8}}$

3. $\frac{5}{6} \times 4$. Write the whole number as $\frac{4}{1}$ so both factors are fractions: $\frac{5}{6} \times \frac{4}{1} = \frac{20}{6}$. Divide top and bottom by 2. $\boxed{\frac{10}{3}}$ (the same as $3\frac{1}{3}$).

4. $\frac{3}{8} \times \frac{2}{9}$. Cancel first: the 3 on top and the 9 below share 3, leaving $\frac{1}{8} \times \frac{2}{3}$; the 2 on top and the 8 below share 2, leaving $\frac{1}{4} \times \frac{1}{3}$. Now multiply across. $\boxed{\frac{1}{12}}$

5. $2\frac{1}{2} \times \frac{3}{5}$. A mixed number must become an improper fraction first: $2\frac{1}{2} = \frac{2 \times 2 + 1}{2} = \frac{5}{2}$. Then $\frac{5}{2} \times \frac{3}{5}$ — cancel the 5s — $= \frac{3}{2}$. $\boxed{\frac{3}{2}}$

6. $\frac{3}{4} \times x = \frac{9}{20}$. The missing factor is found by dividing: $x = \frac{9}{20} \div \frac{3}{4} = \frac{9}{20} \times \frac{4}{3}$. Cancel 3 into 9 and 4 into 20: $\frac{3}{5} \times \frac{1}{1}$. Check: $\frac{3}{4} \times \frac{3}{5} = \frac{9}{20}$ ✓ $\boxed{x = \frac{3}{5}}$

7. $1\frac{1}{3} \times 2\frac{1}{4}$. Convert both: $1\frac{1}{3} = \frac{4}{3}$ and $2\frac{1}{4} = \frac{9}{4}$. Multiply: $\frac{4}{3} \times \frac{9}{4}$. The 4s cancel and 3 divides into 9, leaving $\frac{1}{1} \times \frac{3}{1}$. 3

8. 12 batches, each using $\frac{3}{8}$ of a cup. “Each” with a fraction means multiply: $12 \times \frac{3}{8} = \frac{12}{1} \times \frac{3}{8} = \frac{36}{8}$. Divide top and bottom by 4. $\frac{9}{2}$ cups (that is $4\frac{1}{2}$ cups).

9. $\frac{7}{10} \times \frac{5}{14}$. Cancel before multiplying: 7 divides into 14, and 5 divides into 10, leaving $\frac{1}{2} \times \frac{1}{2}$. Multiplying across gives the answer. $\frac{1}{4}$

10. Area of a rectangle is length times width, so the area is $2\frac{3}{4} \times 1\frac{1}{2}$. Convert both to improper fractions: $\frac{11}{4} \times \frac{3}{2} = \frac{33}{8}$. As a decimal, $33 \div 8 = 4.125$. 4.125 ft^2
 Watch for the common error: multiplying the whole parts and the fraction parts separately gives 2.375, which is far too small — the rectangle is wider than 2 ft and taller than 1 ft, so its area must exceed 2 square feet by more than that.

11. $x \times 2\frac{1}{2} = 4$. Rewrite the mixed number: $2\frac{1}{2} = \frac{5}{2}$, so $x \times \frac{5}{2} = 4$. Divide by $\frac{5}{2}$, which means multiplying by its reciprocal: $x = 4 \times \frac{2}{5} = \frac{8}{5}$. Check: $\frac{8}{5} \times \frac{5}{2} = \frac{40}{10} = 4$ ✓ $x = \frac{8}{5}$

12. Saturday: Maya walks $\frac{2}{3}$ of the trail, so the part left over is $1 - \frac{2}{3} = \frac{1}{3}$ of the trail. The trail is $4\frac{1}{2} = \frac{9}{2}$ kilometres, so the leftover part is $\frac{9}{2} \times \frac{1}{3} = \frac{3}{2}$ km. Sunday she walks half of that leftover piece: $\frac{3}{2} \times \frac{1}{2} = \frac{3}{4}$. $\frac{3}{4} \text{ km}$